CEM AS AN ADAPTIVE MODEL-ERROR AMPLIFIER IN LEARNED WORLD-MODEL MPC

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ABSTRACT

Model-predictive control with learned world models often optimizes imagined returns. Random shooting and Best-of-N select from a fixed proposal, while the cross-entropy method (CEM) repeatedly selects elite imagined rollouts and refits the proposal distribution. We study a narrow failure mode of that adaptation: when learned-model errors create optimistic sparse regions, elite refits can move future samples toward those regions, making model optimism self-reinforcing. We give a simple elite-refit identity showing when pocket mass increases, then test the mechanism in a controlled one-dimensional planning world with a known true optimum, a sparse learned-ensemble variant, and CEM population/iteration sweeps. In the verified full CPU run, vanilla CEM has mean controlled regret 11.64 versus 6.16 for one-shot Best-of-N, while model-side repairs reduce regret to 0.22–0.30 for the strongest variants. In the sparse learned-ensemble experiment, calibrated learned CEM reduces regret from 4.64 to 2.96. In sweeps, combined repair reduces mean CEM regret from 9.08 to 2.33, and short closed-loop return improves from -2.55 to 3.94. These are controlled and synthetic mechanism results, not benchmark-scale or real-robot claims.

1 INTRODUCTION

Learned world models make planning cheap by replacing real interaction with imagined rollouts. This pattern appears in latent CEM planning, latent imagination, uncertainty-aware model-based reinforcement learning, and sampling-based MPC (Hafner et al., 2019; 2020; Chua et al., 2018; Nagabandi et al., 2018; Williams et al., 2017). The optimizer is often treated as a detachable search routine: if the model is accurate enough, stronger search should improve behavior.

This paper isolates a case where that intuition can fail. A static Best-of-N planner can certainly select trajectories that exploit model error, but it samples once from a fixed proposal. CEM samples, scores, selects elites, refits the proposal to those elites, and repeats (Rubinstein, 1999; de Boer et al., 2005; Bharadhwaj et al., 2020). If the elites are selected partly because of model error, the next proposal may move toward the error source. Optimism can become an attractor of the optimizer itself.

Our central claim is deliberately bounded:

> *CEM with a learned model can be worse than a static one-shot proposal because each elite refit changes the proposal toward regions selected by model error, not necessarily real utility.*

We make three contributions. First, we frame CEM's repeated elite refits as an adaptive model-error amplification mechanism. Second, we state a finite elite-refit identity that justifies measuring pocket occupancy, proposal drift, and elite model-error concentration. Third, we report controlled CPU evidence, sparse learned-ensemble evidence, and repair ablations that target the refit mechanism. We do not claim that CEM is generally bad, that uncertainty penalties are new, or that the result transfers to PlaNet/Dreamer-scale benchmarks without further validation.

2 Related Work

World-model planning. PlaNet learns compact latent dynamics and performs online planning with CEM-style shooting (Hafner et al., 2019). Dreamer learns behavior from latent imagination, using gradients through imagined trajectories rather than relying primarily on online CEM (Hafner et al., 2020). PETS combines probabilistic ensembles with trajectory sampling to reduce model-bias damage in MPC (Chua et al., 2018). Earlier neural-dynamics MPC systems and information-theoretic MPC also show the importance of sampled action-sequence optimization (Nagabandi et al., 2018; Williams et al., 2017).

CEM and hybrid MPC. The cross-entropy method is a general adaptive sampling optimizer that repeatedly selects elites and refits a proposal distribution (Rubinstein, 1999; de Boer et al., 2005). Hybrid CEM/gradient MPC improves action-sequence optimization by interleaving sampling and gradient steps (Bharadhwaj et al., 2020). Our focus is not optimizer efficiency. We ask what the elite-refit loop does when the learned objective contains sparse optimism.

Model bias and exploitation. Model-based RL is vulnerable to model error, compounding prediction error, and planner exploitation of learned-model artifacts (Talvitie, 2014; Janner et al., 2019). Prior work motivates uncertainty penalties, ensembles, short rollouts, and conservative model use. The novelty boundary here is narrower: we treat CEM's adaptive refit as the mechanism under audit, and measure whether proposal drift, elite model error, pocket occupancy, and executed regret rise together.

3 Mechanism

Let a denote an action sequence, $U(a)$ the true return, and $S(a)$ the learned-model score. Write model error in return space as $E(a) = S(a) - U(a)$. A one-shot Best-of-N planner draws candidates from a fixed proposal and executes the candidate with maximum S. CEM draws candidates from proposal q_t, selects the top ρ fraction by S, refits q_{t+1} to the elites, and repeats.

Suppose a sparse region of action-sequence space enters a model-error pocket. If pocket trajectories are overrepresented among model-scored elites, CEM's refit moves the next proposal toward the pocket. That movement can increase the chance of sampling more pocket trajectories on the next iteration. The repository measures four primary diagnostics: proposal drift from the initial Gaussian, elite model-error concentration, pocket occupancy among elites, and the selected-tail gap $S(a_{\mathrm{sel}}) - U(a_{\mathrm{sel}})$.

4 Theory

The following identity is intentionally modest. It is not a regret theorem, and diagonal Gaussian CEM only approximates the exact conditional update through elite moment matching. Its purpose is to justify the diagnostic: if a model-error pocket is elite-enriched, proposal drift toward that pocket is the expected behavior of the refit.

Proposition 1 (Elite-refit pocket amplification)**.** *Let $P(a) \in \{0,1\}$ indicate whether action sequence a enters a model-error pocket, and let $L_t(a) \in \{0,1\}$ indicate whether a is elite under proposal q_t. Assume $\Pr_{q_t}(L_t = 1) = \rho$ and an idealized CEM update that refits the next proposal's pocket mass exactly to the elite conditional mass:*

$$q_{t+1}(P = 1) = \Pr_{a \sim q_t}(P(a) = 1 \mid L_t(a) = 1). \tag{1}$$

If $\Pr(L_t = 1 \mid P = 1) > \rho$, then $q_{t+1}(P = 1) > q_t(P = 1)$ and

$$\frac{q_{t+1}(P = 1)}{q_t(P = 1)} = \frac{\Pr(L_t = 1 \mid P = 1)}{\rho}. \tag{2}$$

For any moment-matched trajectory feature $\phi(a)$ under the exact elite-conditional refit,

$$\mathbb{E}_{q_{t+1}}[\phi] - \mathbb{E}_{q_t}[\phi] = \frac{\mathrm{Cov}_{q_t}(\phi, L_t)}{\rho}. \tag{3}$$

Table 1: Main verified evidence. Regret is against the strengthened true-oracle estimate in the controlled world; lower is better. Closed-loop return is higher-is-better.

Setting	Planner or comparison	Metric
Controlled	CEM, pilot calibrated	regret 0.22
Controlled	CEM, disagreement veto	regret 0.23
Controlled	CEM, uncertainty pessimism	regret 0.30
Controlled	CEM, combined repair	regret 0.77
Controlled	Best-of-N	regret 6.16
Controlled	equal-budget Best-of-N	regret 7.99
Controlled	random shooting	regret 9.64
Controlled	vanilla CEM	regret 11.64
Controlled	CEM, diversity floor only	regret 13.31
Learned ensemble	calibrated learned CEM	regret 2.96
Learned ensemble	learned Best-of-N	regret 3.19
Learned ensemble	uncertainty-penalized learned CEM	regret 3.35
Learned ensemble	vanilla learned CEM	regret 4.64
Sweep	combined repair over vanilla CEM	regret $9.08 \rightarrow 2.33$
Closed loop	combined repair over vanilla CEM	return $-2.55 \rightarrow 3.94$

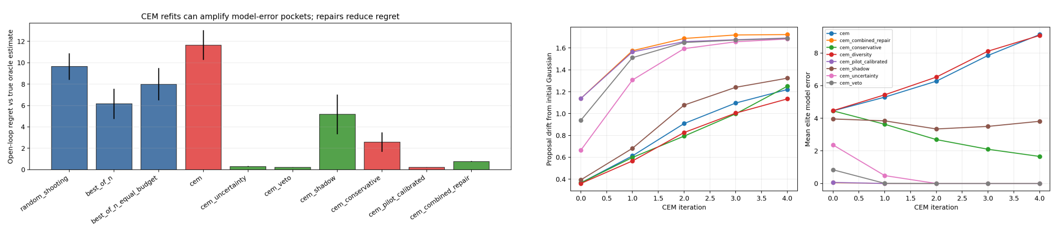


Figure 1: Controlled pocket-world evidence. Left: mean regret by planner shows vanilla CEM underperforming one-shot static proposals while model-side repairs reduce regret. Right: CEM traces expose the mechanism through proposal drift, elite model error, and pocket occupancy rather than only final return.

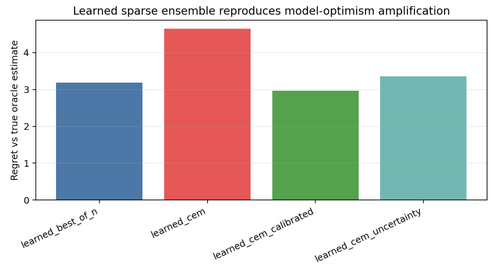
Proof sketch. The pocket-mass identity follows from Bayes' rule: $\Pr(P = 1 \mid L = 1) = \Pr(L = 1 \mid P = 1)\Pr(P = 1)/\Pr(L = 1)$. The feature update follows from the exact conditional distribution, $\mathbb{E}[\phi \mid L = 1] - \mathbb{E}[\phi] = \mathbb{E}[\phi L]/\rho - \mathbb{E}[\phi] = \mathrm{Cov}(\phi, L)/\rho$. Appendix A gives details and limitations.

5 Experiments

Controlled pocket world. The main experiment uses a one-dimensional world with a true goal and a narrow harmful region. The learned model hallucinates positive reward in the harmful sparse region. This makes the true optimum, model-error pocket, and model-versus-true gap inspectable. The verified full CPU run uses 12 seeds, CEM population 256, and 5 CEM iterations. The full-run summary is stored in `results/controlled/controlled_summary.json`.

Sparse learned ensemble. The second experiment trains a bootstrapped polynomial ensemble on transition data with sparse coverage of the harmful region. This checks whether the same diagnostics remain useful when optimism is produced by a learned sparse-region model rather than only by an analytic hallucinated bonus.

Sweeps and closed loop. The sweep varies CEM population, iteration count, and elite fraction over 180 rows. A short receding-horizon evaluation executes the first action, observes the next state, and replans. These artifacts are small CPU experiments; they are intended to stress the mechanism, not to replace benchmark-scale MPC evaluation.



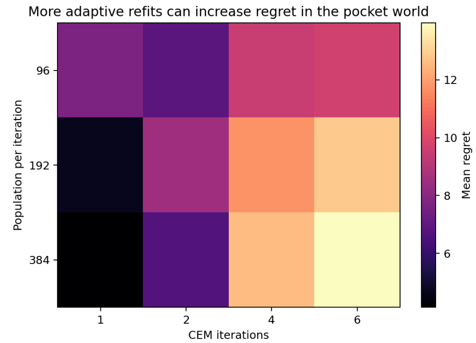


Figure 2: Additional mechanism checks. Left: sparse learned-ensemble regret shows calibrated learned CEM reducing regret relative to vanilla learned CEM. Right: population/iteration/elite-fraction sweeps show where combined repair interrupts high-regret adaptive concentration.

6 Repairs

The repair methods are intentionally simple and auditable. Uncertainty pessimism subtracts ensemble disagreement from the learned score. Model-disagreement veto removes candidates above a disagreement quantile. Pilot-label calibration fits a small optimism penalty using explicitly supplied pilot labels. Conservative refit temperature slows variance collapse. Elite diversity floors prevent all elites from becoming near-duplicates. Shadow-realism scoring penalizes trajectories that look out-of-support under simple support proxies.

These repairs are mechanism probes, not deployable safety guarantees. Tests ensure that repaired planners do not call true evaluation labels during planning. Pilot calibration is the one method that uses real labels, and it uses only labels explicitly supplied by the pilot set.

7 Limitations

The main benchmark is controlled and synthetic. The sparse learned-ensemble experiment is low-dimensional. The proposition is a diagnostic elite-refit identity, not a general regret guarantee. Equal-budget static Best-of-N can also fail when it samples the bad pocket. The current evidence should therefore be read as a mechanism scaffold, not a completed benchmark paper. The next validation step is to port these diagnostics into PlaNet/PETS-style continuous-control benchmarks and test whether latent uncertainty, disagreement, and realism proxies produce the same repair pattern.

8 Conclusion

CEM's strength is adaptation, but adaptation can also amplify the wrong signal. In learned world-model MPC, elite imagined rollouts may be elite because they are truly useful, because they exploit model error, or both. This repository shows a controlled setting where refitting to those elites concentrates search into model-error pockets and damages real return. The repair results suggest that slowing, vetoing, or calibrating uncertain elite concentration is a promising direction, while the final claim remains bounded to controlled CPU evidence.

Reproducibility Statement

The source package includes the ICLR LaTeX files, local style files, copied figures, and the appendix artifact map. Repository claims are gated by `scripts/run_claim_audit.py--repo-root.--claim-filedocs/claims.json`. All main empirical numbers are read from committed CSV/JSON artifacts under `results/`.

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A PROOF DETAILS

The proposition uses an idealized conditional refit. Let $p_t = q_t(P = 1)$ and write $L = L_t$. If the next proposal exactly sets its pocket mass to the elite conditional pocket mass, then

$$\begin{aligned} q_{t+1}(P = 1) &= \Pr(P = 1 \mid L = 1) && (4) \\ &= \frac{\Pr(L = 1 \mid P = 1)\Pr(P = 1)}{\Pr(L = 1)} && (5) \\ &= p_t \frac{\Pr(L = 1 \mid P = 1)}{\rho}. && (6) \end{aligned}$$

Therefore the multiplicative lift is $\Pr(L = 1 \mid P = 1)/\rho$. Pocket mass increases exactly when the pocket is elite-enriched, i.e. $\Pr(L = 1 \mid P = 1) > \rho$.

For a moment-matched feature ϕ under an exact elite-conditional update,

$$\begin{aligned} \mathbb{E}_{q_{t+1}}[\phi] - \mathbb{E}_{q_t}[\phi] &= \mathbb{E}_{q_t}[\phi \mid L = 1] - \mathbb{E}_{q_t}[\phi] && (7) \\ &= \frac{\mathbb{E}_{q_t}[\phi L]}{\rho} - \mathbb{E}_{q_t}[\phi] && (8) \\ &= \frac{\mathbb{E}_{q_t}[\phi L] - \mathbb{E}_{q_t}[\phi]\mathbb{E}_{q_t}[L]}{\rho} && (9) \\ &= \frac{\mathrm{Cov}_{q_t}(\phi, L)}{\rho}. && (10) \end{aligned}$$

Diagonal Gaussian CEM does not represent arbitrary pocket indicators exactly. In the implemented planner, the practical analog is a finite-sample mean and variance update on elite action sequences. This is why the paper treats the identity as a diagnostic law rather than a regret theorem.

B Experiment Settings

Controlled pocket world. The full controlled run uses seeds 0 through 11, CEM population 256, elite fraction 0.1, 5 CEM iterations, horizon 8, action dimension 1, action range $[-1, 1]$, initial standard deviation 0.82, minimum standard deviation 0.035, and zero momentum. It compares random shooting, one-shot Best-of-N, equal-budget Best-of-N, vanilla CEM, uncertainty pessimism, elite diversity floor, model-disagreement veto, pilot-label calibration, conservative refit temperature, shadow-realism scoring, and combined repaired CEM.

Sparse learned ensemble. The learned-ensemble run uses six seeds and trains a small bootstrapped polynomial ensemble on sparse transition coverage around the harmful region. It compares vanilla learned CEM, calibrated learned CEM, uncertainty-penalized learned CEM, and learned Best-of-N. The result is intended to check whether the controlled mechanism appears when optimism comes from learned sparse-region bias.

Sweeps and closed loop. The sweep covers population in $\{96, 192, 384\}$, CEM iterations in $\{1, 2, 4, 6\}$, elite fraction in $\{0.05, 0.1, 0.2\}$, and seeds 0 through 4, for 180 rows. Closed-loop evaluation executes the first action of the selected plan, observes the next state, and replans over a short horizon. The closed-loop result is reported as a small receding-horizon stress test, not as a global theorem.

C Repair Definitions

Uncertainty pessimism. The score subtracts an ensemble-disagreement penalty. This directly targets optimistic candidates in regions where the learned ensemble disagrees.

Model-disagreement veto. Candidates above a disagreement quantile are removed before elite selection. This is stricter than pessimism and can block a high-scoring pocket candidate before it enters the elite set.

Pilot-label calibration. A small explicitly supplied pilot set with true labels fits an optimism penalty. This method spends real labels; the paper does not present it as label-free.

Conservative refit temperature. The refit keeps proposal variance from collapsing too quickly. This targets the adaptive-concentration mechanism rather than the score directly.

Elite diversity floor. The elite set is constrained so that near-duplicate candidates cannot monopolize the refit. The full run shows that this repair alone is not sufficient in the controlled pocket world.

Shadow-realism scoring. The score includes simple support proxies that penalize trajectories that look out-of-support. This is a mechanism probe for OOD elite concentration.

D Artifact Map

E Claim Audit

The repository claim file contains four claim families. Three are supported or partial with evidence files: adaptive elite-refit amplification, CEM underperforming static Best-of-N in the controlled pocket world, and repair variants reducing controlled regret. The benchmark-scale PlaNet/Dreamer transfer claim is explicitly marked unsupported and is not promoted in the main text.

The audit command used for verification is:

```
python scripts/run_claim_audit.py --repo-root .
       --claim-file docs/claims.json
```

Purpose	Repo-relative artifact
Controlled summary numbers	`results/controlled/controlled_summary.json`
Controlled rollouts	`results/controlled/controlled_rollouts.csv`
Controlled CEM traces	`results/controlled/controlled_traces.csv`
Closed-loop returns	`results/controlled/closed_loop.csv`
Learned-ensemble summary	`results/learned_ensemble/learned_summary.json`
Learned-ensemble traces	`results/learned_ensemble/learned_traces.csv`
Sweep repair gain	`results/sweeps/repair_gain.csv`
Sweep raw rows	`results/sweeps/cem_sweep.csv`
Claim gate	`docs/claims.json`, `scripts/run_claim_audit.py`

Table 2: Primary artifacts used by the paper.

The script checks that supported and partial claims cite existing evidence files, unsupported claims do not cite evidence as if proven, and repository prose avoids banned universal language such as "CEM always hurts" or "guarantees improved real return."